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Supplementary Materials for

The translational potential of drug-induced hypothermia in acute ischemic stroke

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Data files S1 to S4
MDAR Reproducibility Checklist

Supplementary Materials and Methods

Animals

The C57BL/6J mice were purchased from SiPeiFu Experimental Animal Co., Ltd.. Mice were housed under a 12h light/dark cycle at a room temperature of 24°C. Food and water intake were not restricted. Mice were randomly assigned to each group, and all experiments were performed under standard laboratory procedures of randomization and blinding. A total of 8 male adult rhesus monkeys (*Macaca mulatta*), aged 8-12 years old and weighted 7.5-9.8 kg, were involved in the non-human primate studies. All monkeys were screened for and were free of tuberculosis, Shigella, Salmonella, helminths, ectoparasites, Entamoeba histolytica, and herpes B virus. Monkeys were housed separately in stainless steel cages and fed with commercially prepared monkey food. Water supply was not restricted. All animal studies were approved by the Institutional Animal Care and Use Committee of Capital Medical University (No. XW-2022060-1 for mice and No. AEEI-2020-038 for rhesus monkeys). All experimental procedures were performed under the National Institutes of Health guidelines for the Care and Use of Laboratory Animals.

Drug dose conversion and administration

Chlorpromazine hydrochloride and promethazine hydrochloride were purchased from Shanghai Harvest Pharmaceutical CO., LTD. The concentration of the stock solution was 25 mg/mL. Before administration in mice, the drugs were diluted at a ratio of 1:1 with saline to a concentration of 2.5 mg/mL. Mice were injected with C+P at a dose of 11.2 mg/kg through intraperitoneal injection as previously described (*30*). Body surface area (BSA) of different species was calculated using the Meeh-Rubner formula (*48*),

$$BSA = kM^{2/3}$$

k is a constant that varies across species, and *M* refers to body mass. Human BSA was calculated using Stevenson formula (*55*),

$$BSA = 0.0061 \times H + 0.0128 \times M - 0.1529$$

H refers to height and *M* refers to body mass. The equivalent dose was converted using following formula,

$$D_b = \frac{D_a \times M_a}{BSA_a} \times \frac{BSA_b}{M_b}$$

After conversion, the equivalent dose that should be used in rhesus monkeys was 2 mg/kg. Specifically, 25 mg/mL C+P was diluted in 50 mL saline. Our primary concern was to explore whether C+P treatment through i.v. route was safe. Thus, C+P was given to the monkeys intravenously within 1 hour under general anesthesia. 6 mg/kg Phenylephrine was injected intraperitoneally 5 min after C+P treatment. For mice receiving 5'-AMP, 150 mg/kg 5'-AMP was injected i.p. .

Temperature monitoring

Core temperature was recorded with a Thermalert Model TH-8 Temperature Monitor (Physitemp Instruments INC., U.S.A) by placing a probe in the rectum of mice and rhesus monkeys. Surface temperature was recorded using an infrared thermal camera (UNI-T Technology Co. Ltd., China). Temperature was calculated automatically by software supplied by the company.

Respiratory flow recording

Respiratory flow was recorded using an OxyletPro metabolic chamber (Panlab, Harvard Apparatus, Spain). In brief, mice were housed in a chamber which had airflow of approximately 400 ml/min. This chamber could automatically record oxygen consumption and CO_2 production for 24h after drug or saline injection. Food and water supply was not limited. For mice underwent middle cerebral artery occlusion (MCAO), they were housed in the chamber for 1 hour to record baseline respiratory flow. Energy expenditure and respiratory quotient were calculated automatically by Metabolism Software v3.0 based on oxygen consumption and CO_2 production.

Middle cerebral artery occlusion in mice

Transient focal cerebral ischemia was induced by right middle cerebral artery occlusion. Mice (21-23g) were anesthetized with isoflurane. After making an incision in the midline, the right common carotid artery (CCA), internal carotid artery (ICA), and external carotid artery (ECA) were exposed. ECA was gently separated and ligated. An incision was made near the proximal end of the separated ECA. A suture (RWD, China) was inserted into the ECA and advanced to the CCA. The suture was then flipped into the ICA and advanced to the MCA. Transient MCAO was maintained for 80 min, followed by removal of the suture. Further assessments were performed 24 h after the surgery. C+P or saline was given immediately after occlusion.

Mouse infarct volume assessment

Infarct volume was assessed using 2,3,5-Triphenyltetrazolium chloride (TTC) staining. 2% TTC was dissolved in 0.2 M PB solution. After euthanasia, the mouse brain was removed and cut into 6 slices. Brain slices were immersed in TTC solution and incubated at 37 °C for 10 min. Infarct volume was calculated as (contralateral hemisphere - ipsilateral tissue with staining)/contralateral hemisphere to avoid the influence of brain edema.

Infusion of [U-$^{13}C_6$]-glucose

Polyethylene catheters (PE-10) were placed into the right jugular vein under isoflurane anesthesia. PE-10 catheters were filled with saline and sealed with a firelighter to avoid air bubbles. After the surgery, mice recovered consciousness. After 4h fast, mice were infused with [U-$^{13}C_6$]-glucose (Cambridge Isotope Laboratories, USA) through the catheter in a conscious state. A bolus of 0.3 mg/g (in 0.3 mL of saline) was infused over 1 min, followed by a rate of 6.9 mg/kg/min (in 0.45 mL of saline) at 150 μL/h for 180 min for each mouse. Mice were euthanized immediately after infusion, and mouse tissues (brain, BAT) were dissected in one minute. Tissues were frozen in liquid nitrogen and stored in a -80 °C refrigerator.

Metabolic flux analysis

15 mg tissue was transferred to a 2 mL microcentrifuge tube containing several steel beads. 200 μL 80% methanol pre-chilled at -80 °C was added, followed by 5 μL of 0.75 mg/mL myristic acid-d27. The sample was grinded and homogenized at 4 °C. After homogenization, an additional 800 μL of pre-cooled 80% methanol was added, and the mixture was briefly vortexed. The tube was then subjected to ultrasonication at 4 °C for 1 minute. The sample was then quenched at -80 °C for 20 minutes. Thereafter, a second ultrasonication step was performed at 4 °C for 3 minutes to maximize solubilization. Cellular debris was pelleted by centrifugation at 13,000 rpm (4 °C) for 20 minutes, after which the entire supernatant was carefully transferred to a new 2 mL tube. The supernatant was dried completely using a centrifugal vacuum concentrator to remove residual solvents. The dried metabolites were dissolved in 45 μL of methoxyamine hydrochloride solution (20 mg/mL in pyridine) (Sigma, USA) and incubated at 37 °C in a metal bath for 90 minutes. Subsequently, 70 μL of MTBSTFA containing 1% t-BDMCS9(Sigma, USA) was added, and the reaction mixture was incubated again at 37 °C for 90 minutes. To remove insoluble particles, the derivatized sample was centrifuged at 10,000 rpm (10 °C) for 10 minutes. Finally, the supernatant was transferred to a glass GC vial fitted with a 250 μL glass insert and analyzed immediately by gas chromatography-mass spectrometry (GC-MS). GC-MS was performed using Agilent 8890 GC/Pegasus BT TOFMS (Leco, USA).

Plasma metabolites detection

Plasma samples from rhesus monkeys were collected after general anesthesia to establish baseline data. A second plasma collection was performed 1.5 hours after drug administration. For patients, plasma collection was carried out at 24 ± 12 hours post-surgery. Metabolite concentrations were measured using following kits: lactate (S0227S, Beyotime, China), pyruvate (S0299S, Beyotime, China), free fatty acid (S0215S, Beyotime, China), and β-hydroxybutyrate (E-BC-K785-M, Elabscience, China). Detailed instructions were provided by the manufacturers.

Rhesus monkey ischemia/reperfusion model

The rhesus monkey ischemia/reperfusion model used in this study was previously described by us (*53, 86*). In brief, ischemia was reached using autologous thrombus to occlude the M2 segment of the right middle cerebral artery. Monkeys were fasted for 12h before the surgery. General anesthesia was induced with Ketamine (10 mg/kg, i.m.) and maintained intravenously with propofol (300 μg kg^{-1} min^{-1}). A single thrombus (10 cm) was injected at the branching site of the right M2 segment of MCA through a microcatheter under the guidance of DSA, and the microcatheter was placed at the proximal part of the MCA. Artery occlusion was confirmed with DSA. After 2h occlusion, t-PA was infused locally through the micro catheter at a dose of 1.1 mg/kg. The effect of thrombolysis was also confirmed with DSA. During the surgery, C+P or saline was infused intravenously immediately after the artery occlusion.

Magnetic resonance Scanning

Infarct volume was investigated using MRI in this study, as we previously described (*87*). MRI scanning was performed on a Magnetom Trio MRI Scanner (3.0 T; Siemens Medical Solutions, Germany). T2 and DWI sequences were used to calculate infarct volumes. MRI scanning was performed at 4h after ischemia and at 1d and 30d. Infarct volumes were calculated using the ITK-Snap contouring software with average diffusion images stacked to reconstruct the infarct in three dimensions. Two independent and blinded observers analyzed the MRI images. Mouse ^{1}H magnetic resonance spectroscopy (MRS) was obtained by using a 7T MRI scanner (BioSpec 70/30 USR, Bruker, Germany). Mice were anesthetized with isoflurane. Firstly, a T2-weighted sequence was scanned as a localizer. Then, single voxel MRS was obtained using 1H-PRESS sequence, TE/TR= 16.5/2500 msecs in combination with VAPOR water suppression. The voxel size was set as 2*1*2 mm and centered at the striatum. Postprocessing was performed on Topspin (Bruker). The relative content of lactate was obtained by integrating its peak and normalizing it to the peak area of total creatine. Rhesus monkey ^{1}H MRS was obtained by using a 3T MRI scanner (SIGNA, General Electric Company, U.S.A). Rhesus monkeys were anesthetized with isoflurane. The single voxel MRS was scanned with TE/TR=35/1500 msecs. The voxel volume was set at 1*1*1 cm. Postprocessing and quantification was performed on MATLAB-based Osprey (*88*). Osprey could automatically split and quantify peaks of different metabolites.

Neurological assessments

Neurological functions were investigated at 1d, 3d, and 7d after surgery using NHPSS, which ranges from 0 (normal) to 41 (severe neurological impairment) (*89*). Total time required for monkeys to pick up food and withdraw using the affected limb was recorded at 14d and 30d to assess motor function (*90*).

Phase I Trial design and overview

The CHILL trial was a phase 1, single-center, prospective, double-blinded, randomized, dose-escalation trial that aimed to define the optimal dose of C+P for AIS. The trial was conducted at Linyi People's Hospital, Shandong Province, China. The trial was registered with ClinicalTrials.gov (NCT06663631). This trial was approved by the institutional review board at Linyi People's Hospital. A committee composed of independent researchers, physicians, and statisticians supervised the design and implementation of the trial.

Participants

We included 32 patients aged between 18 to 80 years old from November 2024 to February 2025. Eligibility criteria included ischemic stroke with a National Institutes of Health Stroke Scale (NIHSS) score ranging from 6-20; time from last known to be well to randomization within 24h; pre-stroke Modified Rankin Scale scoring 0-1; with indications of reperfusion therapy (including intravenous thrombolysis and endovascular thrombectomy) and signed informed consent. All patients received computer tomography (CT) after admission. Eligibility criteria based on CT included CT angiography (CTA) confirmed large vessel occlusion of anterior circulation; had an Alberta Stroke Program Early Computed Tomography Score (ASPECT) score of 6-10; initial infarct volume on CT perfusion (CTP) lesser than 70mL; a ratio of hypoperfused volume to infarcted volume greater than 1.8; absolute mismatch volume greater than 15 mL according to DEFUSE-3 trial (*91*).

Patients were not eligible if they were diagnosed with intracranial hemorrhage (parenchymal hemorrhage and subarachnoid hemorrhage), epilepsy, phenothiazine allergy or contraindication; Iodine contrast medium allergy; baseline blood glucose less than 50mg/dL or greater than 400mg/dL. Exclusion criteria based on CT included a suspicious aortic dissection, multiple infarcts, intracranial tumor.

Randomization and dose-escalation

Participants were randomly assigned 6:2 to receive C+P treatment or placebo. We evaluated C+P doses ranging from 10 mg to 100 mg in four cohorts (10, 20, 50, and 100 mg, respectively). Of note, a dose of 50 mg in humans is approximately equivalent to 2 mg/kg in monkeys. 2 sentinels were set at the 100mg dose and observed for at least 72h before the remaining participants were dozed, provided there were no safety or tolerability concerns as assessed by the investigator. Enrollment into the C+P cohorts was sequential. Subsequent cohorts were dosed after the dosing regimen in the preceding cohort was recommended to be safe and well tolerated based on the safety and laboratory data through at least day 7.

Blinding

Due to the practical challenges of blinding during drug preparation, unblinded nurses or pharmacists will be responsible for preparing the drugs. These unblinded personnel will not participate in patient screening, follow-up, or evaluation. All other study personnel, including patients, surgeons, clinical evaluators, nurses, and imaging physicians, will remain blinded to the patient's group assignment to ensure the objectivity of outcome assessments.

Intervention

All patients enrolled were eligible for EVT. Some patients were given IVT before EVT according to their indications. All patients received general anesthesia during EVT. C+P of each dose was previously diluted into 50 mL saline. 50 mL saline without C+P was given as a placebo. C+P was given intravenously after randomization and administered for 12h at a velocity of 4 mL/h.

Outcomes

The primary object of this study is to determine the safety of C+P treatment in Patients with AIS receiving reperfusion therapy. The primary safety outcome was set as the incidence and severity of all adverse events (AEs). AEs including: 1. Severe hypothermia with body temperature<32 °C 2. Severe hypotension with systolic blood pressure < 90 mmHg that needs additional support 3. Any form of intracranial hemorrhage 4. Coma 5. Death within 72h 6. Respiratory depression defined as a respiration rate <8 bpm/min 7. Extrapyramidal symptoms include Parkinsonism, dystonia, and dyskinesia. AEs are defined as severe adverse events (SAEs) if severity reaches grade 3-5 according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE) ver. 5.0. Secondary outcomes included death from any cause within 90 days; favorable outcomes at 90 days (defined as mRS 0-2); and penumbra decrease at 72 h. Penumbra was defined as the volume of tissue with time to maximum of residue function (Tmax) exceeding 6 seconds while relative cerebral blood flow (compared with normal tissue) is still above 30% (*91*).

The clinical assessments were conducted by masked neurologists trained for years. 90 d mRS scores were obtained by face-to-face or telephone interviews. Penumbra was calculated automatically using the software BioMind.

Plasma sampling

Patients' blood was collected using anticoagulant tubes 24±12 h after drug delivery. Blood was centrifuged immediately after collection to prevent hemolysis. The velocity was 3000 r/min for 10 mins. Dissected plasma was stored in a -80 °C refrigerator.

Plasma sample pre-treatment

The plasma samples were processed using the OmniProt Kit (Westlake Omics, China) with the following procedure. 15 µL of the plasma sample was transferred into a 1.5 mL Eppendorf tube and centrifuged at 4000 rpm for 2 minutes to get the supernatant. Then, 75 µL of Buffer B from the OmniProt kit was added. Subsequently, 90 μL of the OmniProt material working solution was added, and the mixture was incubated at 32 °C with 220 rpm on an incubator shaker for 1 hour. After incubation, the mixture was centrifuged at 17000 rpm for 10 minutes to remove the supernatant. Finally, 0.8 mL of Buffer C from the kit was then added, followed by another centrifugation at 17000 rpm for 10 minutes to remove the supernatant again.

Metabolite Extraction

Transfer 100 μL of the sample into an EP tube, then add 400 μL of extraction solvent and vortex mix for 30 seconds. Sonicate the mixture for 10 minutes in an ice-water bath. Allow the sample to stand at -40 °C for 1 hour. Centrifuge the sample at 4 °C, 12000 rpm for 15 minutes. Transfer the supernatant to a sample vial for instrumental analysis. Prepare a Quality Control (QC) sample by mixing equal amounts of supernatant from all samples and analyze this QC sample along with the others.

Protein digestion

The protein-bound material was subjected to subsequent reduction, alkylation, and proteolytic digestion. Initially, 50 μL of lysis buffer (containing 6 M urea and 2 M thiourea) was added to each tube, followed by incubation at 32 ˚C with 220 rpm for 30 minutes. Subsequently, 4 μL of 200 mM tris(2-carboxyethyl) phosphine (TCEP) was added to each tube, with further incubation at 32 ˚C with 220 rpm for 30 minutes. This was followed by the addition of 4 μL of 800 mM iodoacetamide (IAA) and incubation under the same conditions for another 30 minutes. For proteolysis, 180 μL of 100 mM TEAB was added to each tube, along with 2 μL of trypsin (0.5 μg/μL), and the mixture was incubated at 32 ˚C with 220 rpm for 16 hours. To terminate the digestion, the mixture was centrifuged at 12,000 rpm for 30 seconds, and the supernatant was transferred to a new 1.5 mL Eppendorf tube. Digestion was halted by adding 10% trifluoroacetic acid (TFA) solution, achieving a final TFA concentration of 1%. Confirm that the pH of the samples was between 2 and 3. The SOLAμTM SPE Plate (Thermo Fisher Scientific, USA) was applied for desalting.

Proteome sample analysis

Liquid chromatography-mass spectrometry (LC-MS) analysis was performed using a Vanquish Neo UHPLC system coupled to an Orbitrap Astral mass spectrometer (Thermo Scientific, San Jose, USA) for data-independent acquisition (DIA) analysis. The mobile phase A consisted of 98% water with 0.1% formic acid and 2% Acetonitrile. Mobile phase B comprised 20% water with 80% acetonitrile and 0.1% formic acid. All reagents were of MS grade. For the DIA acquisition, the peptide concentration was set to 0.1 μg/μL, with an injection volume of 4 μL. The amount of sample loaded for each DIA acquisition was 400 ng. During sample acquisition, peptides were loaded onto a pre-column (5 μm, 5 mm*300 μm i.d.) at a pressure of 800 bar, then eluted onto an analytical column (1.9 μm, 120 Å, 150 mm*75 μm i.d.) at a flow rate of

500 nL/min. An 18-minute effective LC gradient (8% to 40% mobile phase B) was used for analysis.

The mass spectrometry scan parameters were configured as follows: FAIMS voltage was set to -42 V. The primary scan was conducted over a range of 380 to 980 m/z, with a resolution of 240,000. The normalized AGC target was set to 500%, and the maximum injection time (max IT) was 3 ms. For secondary scans, the m/z range was adjusted to 150-2000, maintaining the normalized AGC target at 500%. The collision energy was calibrated to 25%, with a maximum injection time of 3 ms. The precursor ion mass range for these scans was specified as 380-980 m/z, with an isolation window of 2 m/z and no overlap between windows. A total of 299 windows were utilized for the analysis.

Mass spectrometry data analysis

The mass spectrometry data were processed using the DIA-NN software (version 1.9) for database searching. Carbamidomethylation of cysteine was set as static modifications, while oxidation of methionine was set as variable modifications. This comprehensive analysis provided both qualitative and quantitative data, applying a stringent threshold for the false discovery rate (FDR) of less than 0.01 as the filtering criterion.

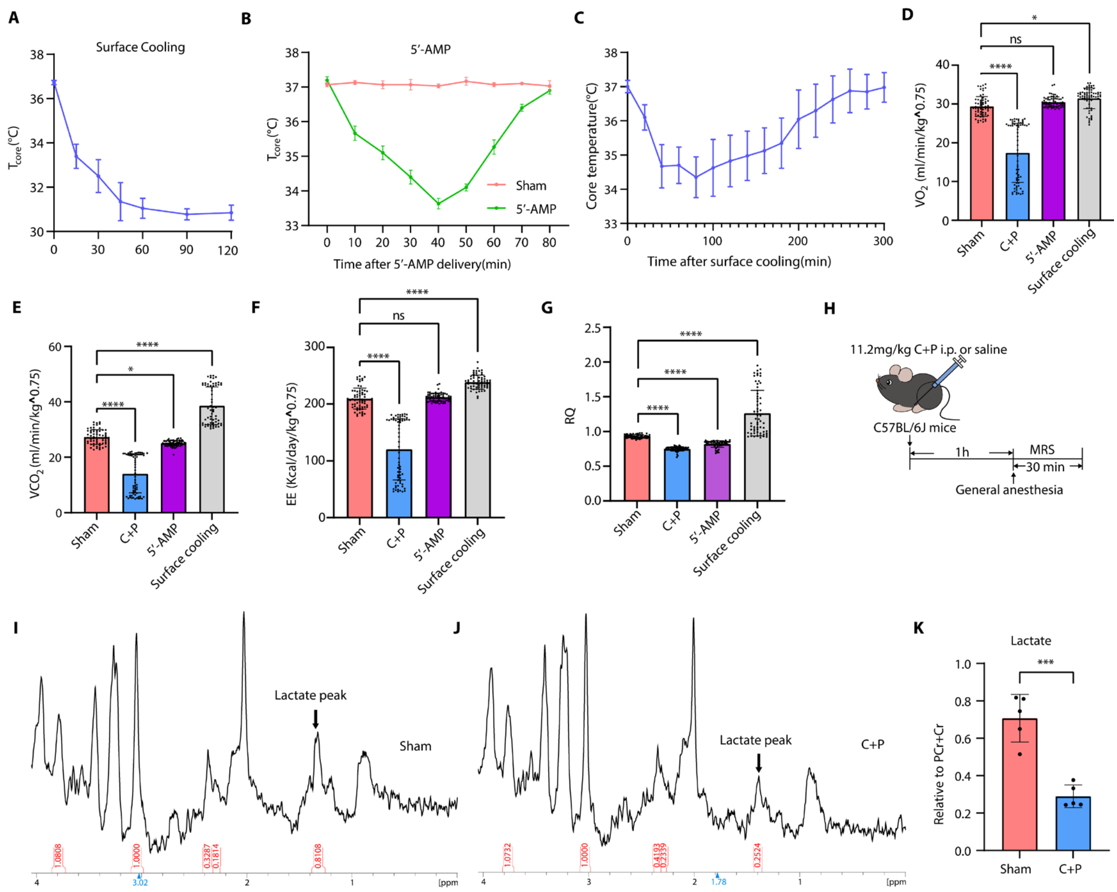

Fig. S1 Assessments of the models in mice. (**A**)Temperature curve of animals receiving surface cooling (n=4). Surface cooling was achieved by covering the mice with ethanol and blowing with a fan. (**B**) Temperature curve of animals receiving 150 mg/kg of 5'-AMP (n=4). **(C)**Temperature curve of animals receiving surface cooling in the metabolic chamber (n=4). Surface cooling was achieved by covering the mice with water and covering the bottom of metabolic chamber with ice packs. **(D)**Quantification of VO_2 in metabolic chamber. Each dot represents the mean VO_2 of the animals at a specific time point. (Kruskal-Wallis test followed by Dunn's multiple comparisons test, n=5). **(E)**Quantification of VCO_2 in metabolic chamber. Each dot represents the mean VCO_2 of the animals at a specific time point. (Kruskal-Wallis test followed by Dunn's multiple comparisons test, n=5). **(F)**Quantification of EE in metabolic chamber. Each dot represents the mean EE of the animals at a specific time point. (Kruskal-Wallis test followed by Dunn's multiple comparisons test, n=5). **(G)**Quantification of RQ in metabolic chamber. Each dot represents the mean RQ of the animals at a specific time point. (Kruskal-Wallis test followed by Dunn's multiple comparisons test, n=5).**(H)** Illustration of experimental paradigm and timeline for MRS scanning in anesthetized animals receiving C+P treatment or saline (n=5). **(I)** Representative image of cerebral magnetic resonance spectroscopy in sham mice. **(J)**Representative image of cerebral magnetic resonance spectroscopy in mice receiving C+P treatment . The relative content of lactate was obtained by integrating its peak and normalizing it to the peak area of total creatine. **(K)** Quantification of lactate content in mouse brain. The relative content of lactate was obtained by integrating its peak and normalizing it to the peak area of total creatine. (unpaired Student's *t*-test, n=5). Data are expressed as mean ± SD. *indicates $P<0.05$, ***indicates $P<0.001$, ****indicates $P<0.0001$. EE, energy expenditure; RQ, respiratory quotient.

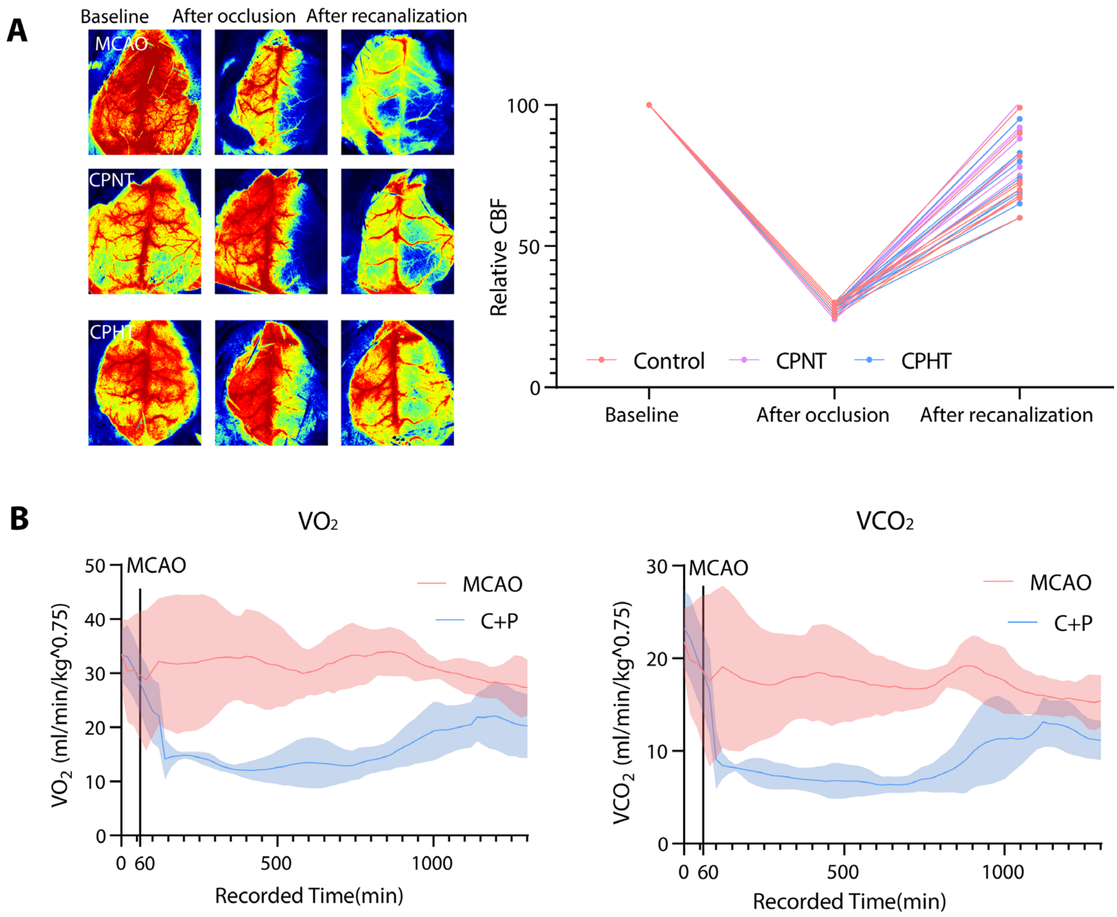

Fig. S2 Confirmation of mouse MCAO model and respiratory parameters after surgery. **(A)** Left, Representative image of laser doppler flowmetry. Right, quantification of CBF before, during, after ischemia(n=10). Ischemic models were considered successful only when relative CBF dropped to 30% compared to the contralateral hemisphere. Data are represented as individual values. **(B)** O_2 consumption and CO_2 production at baseline and after MCAO model (n=4). Data are represented as mean ± SD.

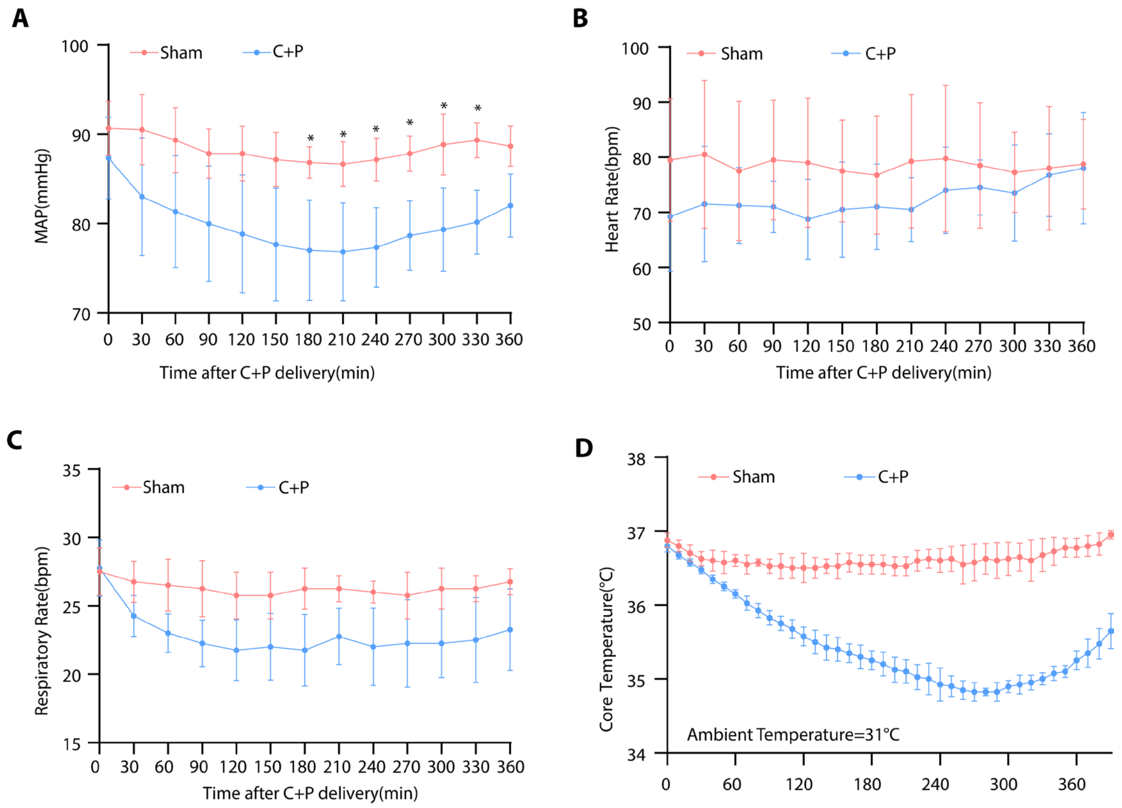

Fig. S3 Vital signs of monkeys receiving C+P treatment. (**A**) Mean arterial pressure after C+P treatment (two-way ANOVA followed by Sidak multiple comparisons test, n=4). (**B**) Heart rate after C+P treatment (n=4). (**C**)Respiratory rate after C+P treatment (n=4). (**D**) Core temperature after C+P administration when ambient temperature was set at 31˚C (n=4). Data are represented as mean ± SD. *, *P*<0.05.

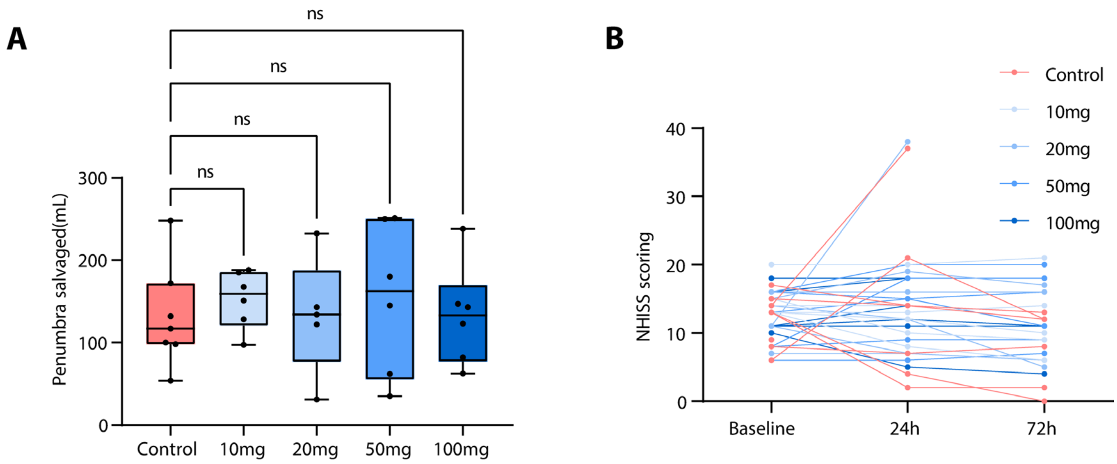

Fig.S4 Penumbra salvaged and NHISS score. (A) Penumbra salvaged at 72h (n_{con}=7, n_{10mg}=6, n_{20mg}=5, n_{50mg}=6, n_{100mg}=6). Data are represented as mean ± SD. ns depict not significant. **(B)** NHISS scores of patients at baseline, 24h and 72h. Data are represented as individual values.

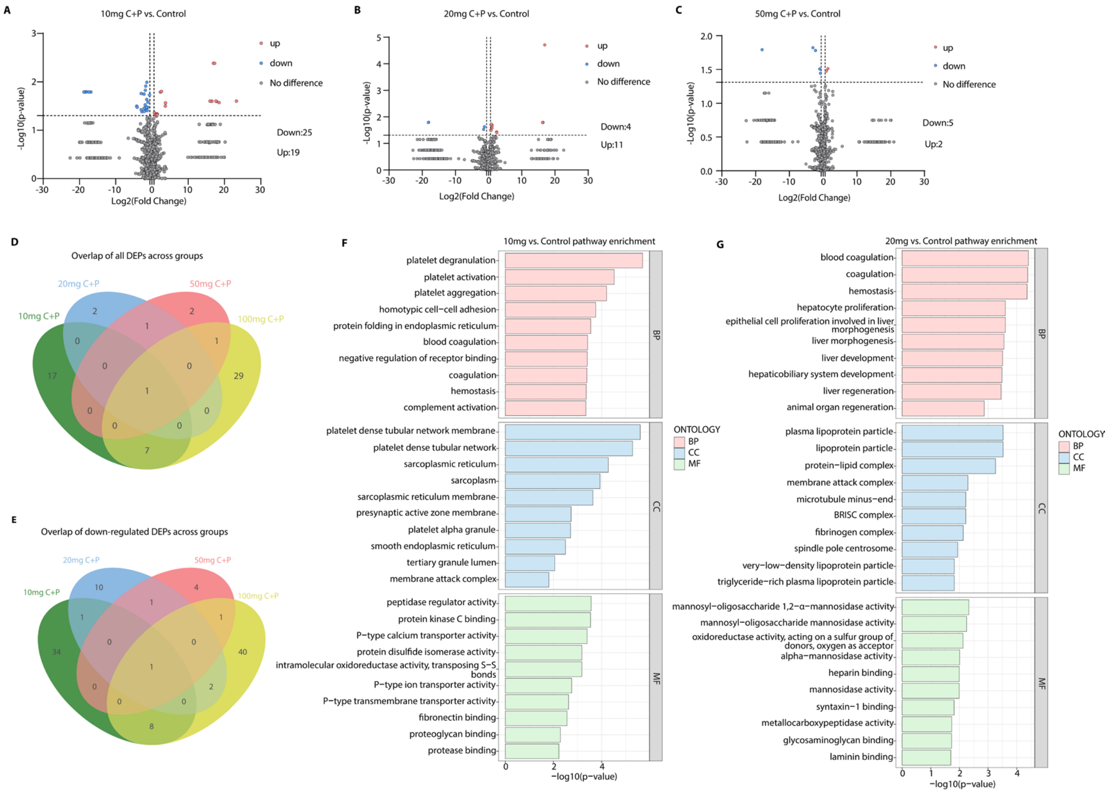

Fig.S5 Proteomic profiling of plasma from patients receiving 10mg, 20mg, and 50mg C+P treatment(**A**)Volcano plot showing differentially represented proteins in plasma of patients treated with 10mg C+P compared to patients treated with saline. X-axis shows fold change and y-axis significance level. Blue dots show significantly downregulated proteins, red dots show significantly upregulated proteins (≥1.5(or ≤-1.5 fold change, P<0.05). (**B**)Volcano plot showing differentially represented proteins in plasma of patients treated with 20mg C+P compared to patients treated with saline. X-axis shows fold change and y-axis significance level. Blue dots show significantly downregulated proteins, red dots show significantly upregulated proteins (≥1.5(or ≤-1.5 fold change, P<0.05). (**C**)Volcano plot showing differentially represented proteins in plasma of patients treated with 50mg C+P compared to patients treated with saline. X-axis shows fold change and y-axis significance level. Blue dots show significantly downregulated proteins, red dots show significantly upregulated proteins (≥1.5(or ≤-1.5 fold change, P<0.05). (**D**)Venn diagram summarizing overlaps of all differentially expressed proteins across groups of different dosages**.** (**E**)Venn diagram summarizing the overlap of downregulated proteins across groups of different dosages. (**F**) Gene Ontology pathway analysis revealing TOP10 enriched pathways in patients' plasma from the 10mg group compared to the control group. (**G**)Gene Ontology pathway analysis revealing TOP10 enriched pathways in patients' plasma from the 20mg group compared to the control group.

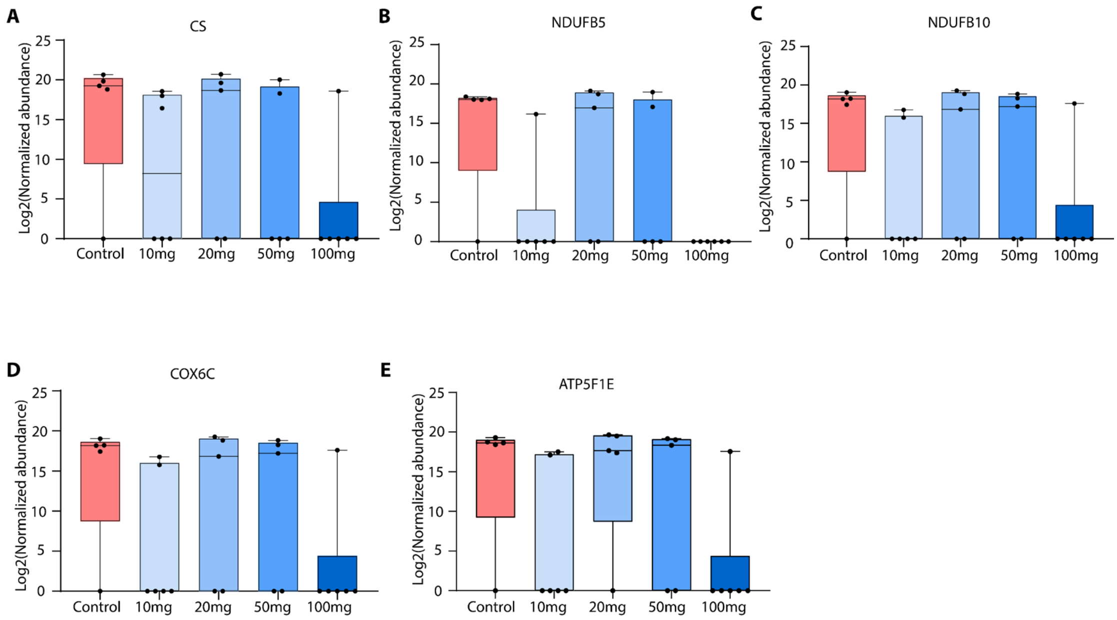

Fig.S6 Expression of proteins associated with aerobic respiration in groups of different C+P doses. (**A-E**) Expression of CS, NDUFB5, NDUFB10, COX6C, and ATP5F1E in different groups ($n_{control}$=5, n_{10mg}=6, n_{20mg}=5, n_{50mg}=5, n_{100mg}=6).

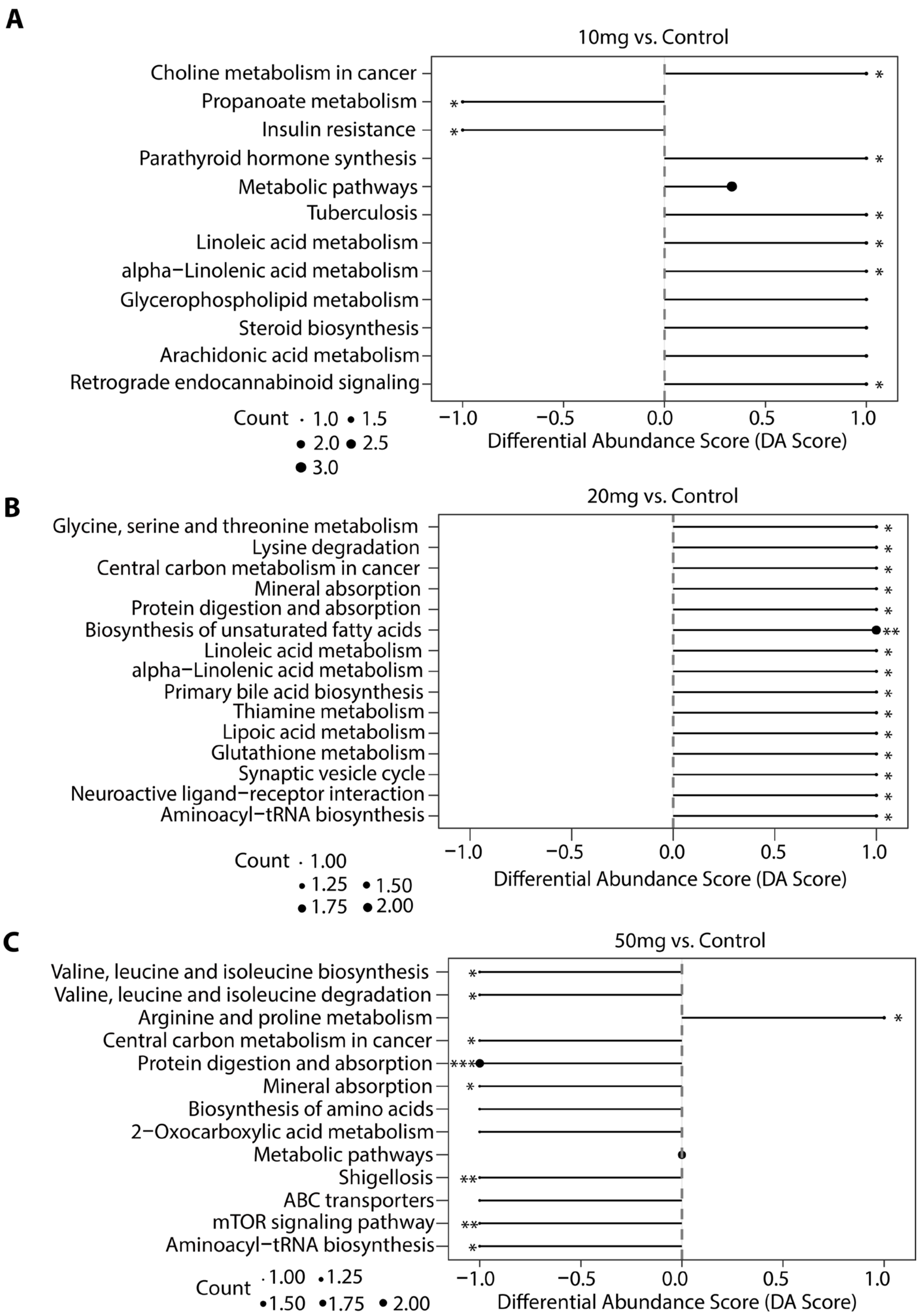

Fig.S7 KEGG pathway analysis of untargeted metabolomics profiling in patients receiving 10/20/50mg C+P compared to saline (**A**) KEGG pathway analysis exhibiting pathways significantly mediated by 10mg C+P. (**B**) KEGG pathway analysis exhibiting pathways significantly mediated by 10mg C+P. (**C**) KEGG pathway analysis exhibiting pathways significantly mediated by 10mg C+P. * indicates P<0.05, **indicates P<0.01, ***indicates P<0.001.

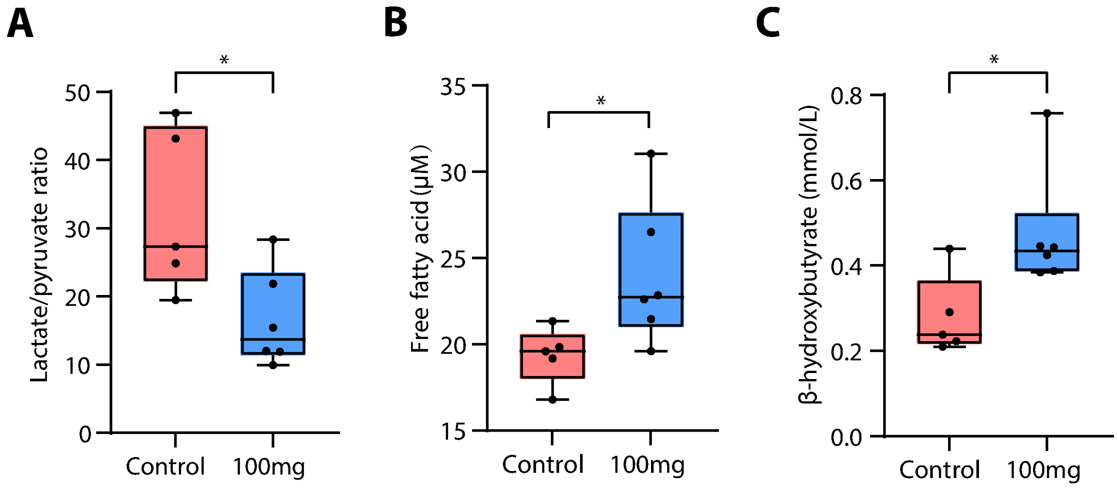

Fig.S8 Plasma metabolites in patients receiving 100mg C+P treatment or saline. (A) Plasma Lactate/pyruvate ratio after 100mg C+P treatment or saline (Welch's *t*-test, $n_{control}$=5, n_{100mg}=6). **(B-C)** Free fatty acid and β-hydroxybutyrate in patients' plasma after 100mg C+P treatment or saline (Welch's *t*-test, $n_{control}$=5, n_{100mg}=6). Data are represented as mean ± SD. *, *P*<0.05.

Table S1 Heart rate within 72h after intervention										
	Con (n = 7) *	10mg(n=6)	*P*-value‡	20mg(n=6)	*P*-value‡	50mg(n=6)	*P*-value‡	100mg(n=6)	*P*-value‡	*P*-value†
1h	75.71±8.18	71.00±9.84	0.8717	70.50±10.03	0.8286	66.83±11.55	0.4453	77.50±15.72	0.9957	0.4820
6h	83.86±18.77	84.00±26.11	0.9999	79.17±5.02	0.9678	85.67±18.33	0.9990	75.00±7.24	0.7698	0.8081
24h	77.43±10.77	78.17±10.52	0.9998	80.67±12.32	0.9437	76.17±7.39	0.9982	73.67±8.21	0.9085	0.8098
36h	79.71±11.93	84.83±14.12	0.8193	77.50±7.58	0.9888	76.83±10.72	0.9710	79.00±8.25	0.9998	0.7297
48h	80.14±9.03	81.67±16.99	0.9985	75.00±6.08	0.8829	79.17±16.40	0.9998	85.00±5.48	0.9017	0.7366
60h	80.00±10.47	87.67±23.42	0.7417	70.67±4.51	0.5982	74.67±14.61	0.9069	80.33±11.31	0.9999	0.3174
72h	83.43±11.31	73.33±9.59	0.1517	72.33±7.73	0.1016	73.83±8.61	0.1835	76.17±4.54	0.4036	0.1698

*Heart rate data was missing for one patient in the control group.

‡*P*-value for post hoc Dunnett's t-test.

†*P*-value for overall one-way ANOVA test.

Table S2 Respiratory rate within 72h after intervention		

	Con (n = 7) *	10mg(n=6)	*P*-value‡	20mg(n=6)	*P*-value‡	50mg(n=6)	*P*-value‡	100mg(n=6)	*P*-value‡	*P*-value†
1h	15.00±1.95	14.67±0.52	0.9976	17.67±2.94	0.1932	16.50±0.55	0.6637	14.83±4.26	0.9998	0.1821
6h	18.00±1.92	17.33±3.67	0.9558	17.83±1.47	0.9998	18.17±1.84	0.9998	17.67±1.21	0.9965	0.9714
24h	18.29±0.95	18.50±1.05	0.9845	17.83±1.17	0.8202	18.17±0.98	0.9982	17.67±0.52	0.6161	0.5636
36h	18.86±1.86	19.00±1.55	0.9987	18.83±0.75	0.9999	17.17±0.75	0.0700	17.33±0.52	0.1149	0.0266
48h	18.14±2.27	18.17±1.64	0.9999	18.83±0.98	0.8352	17.83±1.17	0.9887	18.00±0.89	0.9992	0.8200
60h	17.57±0.79	19.33±1.14	0.0100	18.00±0.89	0.8450	18.00±0.89	0.8450	16.83±1.17	0.4668	0.0026
72h	17.43±1.27	18.17±2.39	0.6998	17.33±0.52	0.9998	18.33±0.52	0.5405	17.33±0.52	0.9998	0.4827

*Respiratory rate data was missing for one patient in the control group.

‡*P*-value for post hoc Dunnett's t-test.

†*P*-value for overall one-way ANOVA test.

Table S3 Detailed Systolic blood pressure of patients.

	Control							
	S001	S007	S011	S016	S018	S023	S026	S032
Baseline	102	166	116	152	130	121	144	132
1h	99	130	122	156	139	142	128	100
6h	122	78	124	154	136	180	122	131
10h	119	Missing Data	127	146	136	152	128	128
15h	126		117	128	134	150	125	130
20h	132		105	130	136	148	129	122
24h	133		120	138	134	147	134	135
36h	115		120	135	135	139	145	138
48h	121		118	119	132	145	143	116
60h	126		102	118	130	141	138	110
72h	122		110	133	132	142	126	118

Table S3 (continued) Detailed Systolic blood pressure.

	10 mg					
	S002	S003	S004	S005	S006	S008
Baseline	159	109	92	110	158	133
1h	130	119	91	98	135	114
6h	135	162	139	99	141	126
10h	123	113	139	99	117	124
15h	133	137	132	97	113	124
20h	134	130	132	91	106	144
24h	126	113	134	102	127	112
36h	105	110	144	106	109	117
48h	102	120	140	122	135	116
60h	112	125	138	116	115	119
72h	122	125	139	110	130	130

Table S3 (continued) Detailed Systolic blood pressure.

	20 mg					
	S009	S010	S012	S013	S014	S015
Baseline	142	155	160	150	163	135
1h	110	110	122	123	140	144
6h	122	131	138	98	118	158
10h	122	139	128	102	133	132
15h	141	139	139	100	118	112
20h	130	130	152	98	128	121
24h	135	94	142	95	125	120
36h	139	101	137	126	133	139
48h	128	112	146	122	118	135
60h	120	139	142	102	128	130
72h	118	130	136	100	126	118

Table S3 (continued) Detailed Systolic blood pressure.

	50 mg					
	S017	S019	S020	S021	S022	S024
Baseline	145	166	167	149	160	140
1h	142	140	146	134	168	128
6h	114	141	152	117	168	150
10h	142	124	129	120	146	102
15h	117	122	130	130	142	100
20h	125	117	129	119	142	120
24h	139	128	134	136	132	123
36h	142	122	132	137	134	96
48h	136	111	147	132	134	130
60h	144	103	145	132	131	126
72h	124	116	148	139	134	118

Table S3 (continued) Detailed Systolic blood pressure.

	100 mg					
	S025	S027	S028	S029	S030	S031
Baseline	133	130	171	100	129	132
1h	116	119	144	104	118	114
6h	113	150	117	131	91	94
10h	110	114	114	124	104	104
15h	126	114	126	124	108	104
20h	126	112	128	120	105	107
24h	134	126	112	116	105	134
36h	136	121	129	111	114	114
48h	126	115	152	126	132	116
60h	134	104	153	125	112	116
72h	135	110	139	141	125	103